Sep 10 Warmups

Find $\int \tan^3(x)\sec(x)\,dx$ using strategy 2.2

$= \int \tan^2(x)\ \boxed{\sec(x)\tan(x)\,dx}$ Hey this looks familiar

$\frac{d}{dx}\sec(x) = \sec(x)\tan(x)$

$dv = \sec(x)\tan(x)\,dx$

$v = \int \sec(x)\tan(x)\,dx$

$v = \sec(x)$

put $\tan^2$ in terms of v

$\tan^2\theta + 1 = \sec^2\theta$

$\tan^2\theta = \sec^2\theta - 1$

$= v^2 - 1$

$\int (v^2 - 1)\,dv$

$= \frac{1}{3}v^3 - v + C = \frac{1}{3}\sec^3(x) - \sec(x) + C$